Algorithms HW

1. Suppose that $f : \mathbb{C} \to \widehat{\mathbb{C}}$ is a meromorphic function with two periods ω_1, ω_2, such that the ratio ω_1/ω_2 is real.

   (a) Suppose ω_1/ω_2 is rational. Show that the periodicity assumption is equivalent to being periodic with respect to some single ω'.

   (b) If ω_1/ω_2 is irrational, show f is constant.

1. Let us write $\omega_1/\omega_2 = p/q$ for some coprime integers p, q. Let us define $\omega_0 := \omega_1/p$. We need to show that (1) f is periodic with repsect to ω_0 and (2) that ω_1, ω_2 are multiples of ω_0.

   We show (2) first. Notice that $\omega_1 = p\omega_0$ by definition. And also, $\omega_2 = \frac{q\omega_1}{p} = q\omega_0$.

   Next we show (1), that f is periodic with repsect to ω_0. Recall Bézout's identity gives us that there exists integers m, n such that $mp + nq = 1$ since p, q are coprime. Rearragnging gives us that

$$\begin{aligned} \frac{\omega_1}{p} &= \omega_1 \left(m + n\frac{q}{p} \right) \\ &= \omega_1 \left(m + n\frac{\omega_2}{\omega_1} \right) \\ \omega_0 &= m\omega_1 + n\omega_2. \end{aligned}$$

   Since f is periodic in ω_1 and ω_2 it thus follows that f is periodic in ω_0.

2. Let $\tau := \omega_1/\omega_2$, we have that τ is irrational. Let $m, n \in \mathbb{Z}$ and notice that $m\omega_1 + n\omega_2 = \omega_2(m\tau + n)$. Now since τ is irrational we have that the set $\{\omega_2(m\tau + n)\}$ is dense in the line $\mathbb{R}\omega_2$. We will show that f is thus constant on the line $\mathbb{R}\omega_2$.

   Let $c \in \mathbb{R}\omega_2$ be a point where f is holomorphic. Since $\{\omega_2(m\tau + n)\}$ is dense in $\mathbb{R}\omega_2$ we have a sequence $z_k = \omega_2(m_k\tau + n_k)$ with $m_k, n_k \in \mathbb{Z}$ which converges to c. Since f is holomorphic at c we have that it is in particular continuous at c, and so $f(c) = \lim_{k\to\infty} f(z_k)$. Moreover since $\omega_2(m_k\tau + n_k) = m_k\omega_1 + n_k\omega_2$ for all k we have that $f(\omega_2(m\tau + n)) = f(0)$ for all $m, n \in \mathbb{Z}$. But then $f(c) = \lim_{k\to\infty} f(z_k) = f(0)$, i.e. f is constant along $\mathbb{R}\omega_2$.

We have shown that f is constant on a closed subset of its domain of holomorphicity. It follows by the identity theorem that f is thus constant on its entire domain.

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2. Given a lattice $\Lambda \subset \mathbb{C}$, addition of complex numbers naturally descends to a group law on the torus $\mathbb{C}/\Lambda$. Let $F : \mathbb{C}/\Lambda \to \widehat{\mathbb{C}}$ be a meromorphic function with zeros at $z_1, \dots, z_n$ and poles at $p_1, \dots, p_m$ (appearing with multiplicity equal to the order). Prove that

$$\sum z_i - \sum p_j = 0,$$

where the addition is with respect to the group law described above.

Since F is a meromorphic function on $\mathbb{C}/\Lambda$ it lifts to a doubly periodic meromorphic function on $\mathbb{C}$, let us also refer to this meromorphic function on $\mathbb{C}$ as F (I will note when we start thinking about it as a function on the torus again). Consider a fundamental paralellogram P of Λ in $\mathbb{C}$ such that F has no zeroes or poles on the boundary of P. Recall that the following variation of the argument principle integral counts the sum of the points which gives zeroes and poles for F.

$$\int_{\partial P} z \cdot \frac{f'(z)}{f(z)} dz = \sum_{z_i \in \mathbb{C}} z_i - \sum_{p_j \in \mathbb{C}} p_j.$$

Let us normalize our lattice so that one of the generators is given by 1 and the other is given by $\tau \in \mathbb{H}$. Let us label the sides of our fundamental paralellogram by $\alpha, \alpha', \beta, \beta'$. Then notice that we have the following identity

$$\begin{aligned}
\int_{\beta} z \cdot \frac{f'(z)}{f(z)} dz &= \int_{-\beta'+\tau} z \cdot \frac{f'(z)}{f(z)} dz \\
&= \int_{-\beta'} (z+\tau) \cdot \frac{f'(z+\tau)}{f(z+\tau)} dz \\
&= \int_{-\beta'} (z+\tau) \cdot \frac{f'(z)}{f(z)} dz \\
&= -\int_{\beta'} (z+\tau) \cdot \frac{f'(z)}{f(z)} dz,
\end{aligned}$$

where the third equality follows since f is periodic in τ and so f' is also periodic in τ. Similar reasoning gives that

$$\int_{\alpha} z \cdot \frac{f'(z)}{f(z)} dz = -\int_{\alpha'} (z+1) \cdot \frac{f'(z)}{f(z)}.$$

Now, consider the following

$$\begin{aligned}\int_{\partial P} z \cdot \frac{f'(z)}{f(z)} dz &= \int_{\alpha+\beta+\alpha'+\beta'} z \cdot \frac{f'(z)}{f(z)} dz \\ &= \int_{\alpha'} \frac{f'(z)}{f(z)} dz - \tau \int_{\beta'} \frac{f'(z)}{f(z)} dz.\end{aligned}$$

Now, we claim (as prompted by the book hint) that the integral $\int f'(z)/f(z)dz$ along any of the side lengths of the fundamental parallelogram is some multiple of $2\pi i$. Consider

$$\int_{\alpha'} \frac{f'(z)}{f(z)} dz = \int_{z_0}^{z_0+\tau} \frac{f'(z)}{f(z)} dz = \log(f(z_0)) - \log(f(z_0+\tau)),$$

where z_0 is one endpoint for α'. Note that, since we are integrating just over an interval in $\mathbb{C}$, we can take the branch cut of log to be away from either of the line segments that we care about. Now recall that f is periodic with respect to τ, i.e., $f(z_0) = f(z_0+\tau)$. That means, the image of f under α' traces out some closed curve in $\mathbb{C}^\times$ (recall that we chose P so that f does not vanish on ∂P). It follows then that

$$\int \frac{f'(z)}{f(z)} dz = k(2\pi i),$$

for some integer k.

Now returning the the integral that we are concerned with, we thus have

$$\int_{\alpha'} \frac{f'(z)}{f(z)} dz - \tau \int_{\beta'} \frac{f'(z)}{f(z)} dz = (2\pi i)(k_1 - k_2\tau).$$

For some $k_1, k_2 \in \mathbb{Z}$. Now notice that $k_1 - k_2\tau$ is a lattice point. And so $k_1 - k_2\tau \equiv 0 \bmod \Lambda$. In other words, we have shown

$$\sum z_i - \sum p_j = \int_{\partial P} z \cdot \frac{f'(z)}{f(z)} dz \equiv 0 \bmod \Lambda.$$

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3. Given a lattice $\Lambda \subset \mathbb{C}$, show that the series

$$\sum_{\omega \in \Lambda^*} \frac{1}{|\omega|^2}$$

is divergent.

Recall that without loss of generality we can take our lattice Λ to be generated by 1 and $\tau = a + bi$ with $a, b \in \mathbb{R}$ and $b > 0$. We can also choose generators of our lattice so that $a > 0$. With this notation we have that $\omega = m + n\tau$ for integers m, n, and so

$$|\omega|^2 = |m + n(a + bi)|^2 = (m + na)^2 + (nb)^2.$$

Our series of interest becomes

$$\sum_{\omega \in \Lambda^*} \frac{1}{|\omega|^2} = \sum_{m,n \in \mathbb{Z}} \frac{1}{(m + na)^2 + (nb)^2}.$$

For m, n not both simultaneously zero. Consider this sum for some fixed n. We can split our sum

$$\begin{aligned}
\sum_{m,n \in \mathbb{Z}} \frac{1}{(m + na)^2 + (nb)^2} &= \sum_{(m+na)^2 \leq (nb)^2} \frac{1}{(m + na)^2 + (nb)^2} + \sum_{(m+na)^2 > (nb)^2} \frac{1}{(m + na)^2 + (nb)^2} \\
&\geq \sum_{(m+na)^2 \leq (nb)^2} \frac{1}{(m + na)^2 + (nb)^2} \\
&\geq \sum_{(m+na)^2 \leq (nb)^2} \frac{1}{2(nb)^2} \\
&= K(n) \cdot \frac{1}{2(nb)^2}.
\end{aligned}$$

Where $K(n)$ is the number of integers m satisfying $(m + na)^2 \leq (nb)^2$. The second line follows since the latter sum on the first line contains only positive terms. The third line follows since, by the condition in the sum, $(nb)^2 \geq (m + na)^2$ for each term. Counting the number of integers m satisfying $(m + na)^2 \leq (nb)^2$ gives us that $K(n) \sim 2\,|nb|$ and so we have

$$\sum_{m,n \in \mathbb{Z}} \frac{1}{(m + na)^2 + (nb)^2} \geq \frac{1}{|nb|}.$$

Now we can show that our series of interest diverges

$$\sum_{\omega\in\lambda}\frac{1}{|\omega|^2}=\sum_{m,n\in\mathbb{Z}}\frac{1}{(m+na)^2+(nb)^2}\geq\sum_{n\in\mathbb{Z}}\frac{1}{|nb|}=\frac{1}{b}\sum_{n\in\mathbb{Z}}\frac{1}{|n|},$$

which diverges.

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